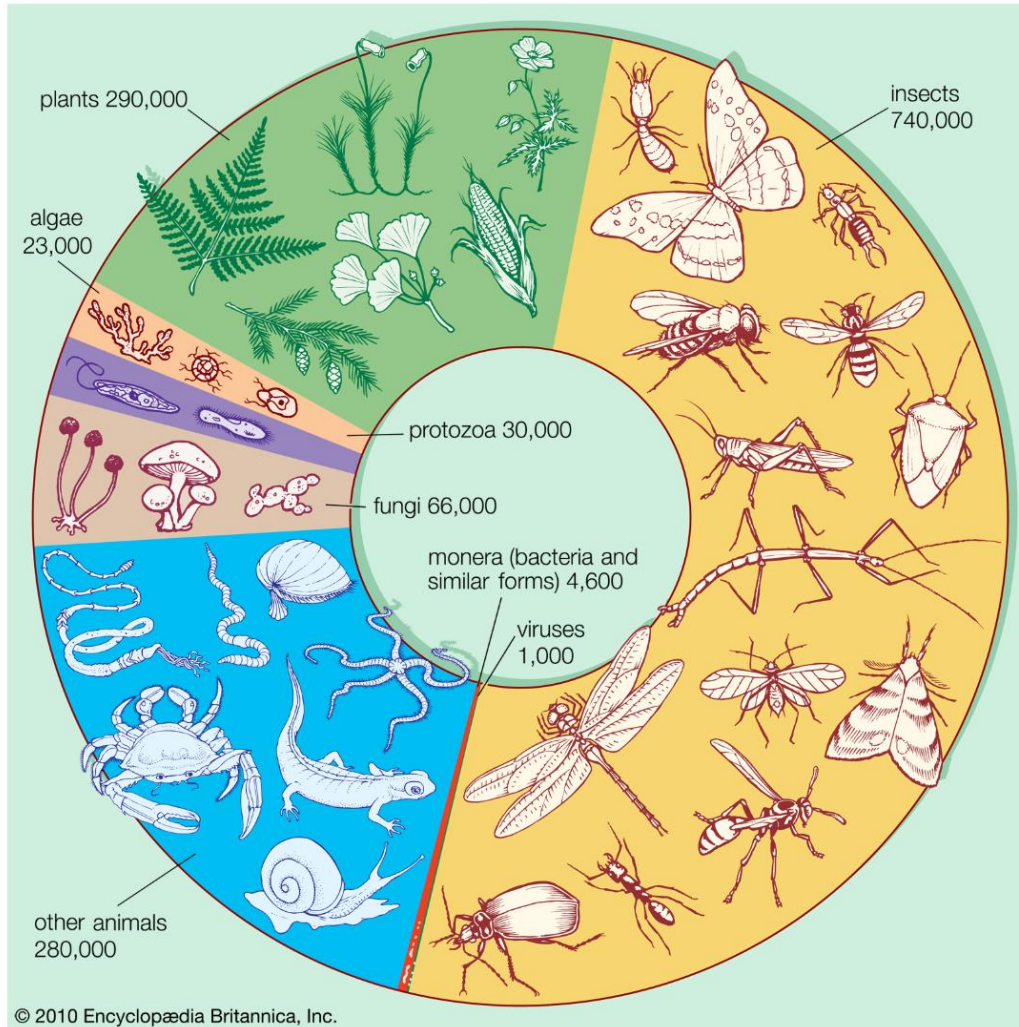


Evolution

By: Hassaan Mehboob Awan, Ph.D.





<https://www.britannica.com/science/biosphere/The-diversity-of-life>

The diversity of life

- All cellular life has dsDNA genome
- All cells are enclosed by lipid membranes
- The genetic code is nearly universal
- Amino acids can occur in L or D forms, but cells use L-shaped amino acids

Nothing in Biology Makes Sense
Except in the Light of Evolution
Dobzhansky (1973)





Dr. Peter Grant Dr. Rosemary Grant

Emeritus Professors of Ecology and Evolutionary Biology, Princeton University



Formation of new lineage of finches – The Big Bird lineage



*Rapid hybrid speciation in Darwin's finches. **Science, 2017***





Direct documentation from origin to reproductive isolation



Interbreeding of two species may result in the formation of a new specie



Reproductive isolation, which typically develops over hundreds of generations, can be established only in three.





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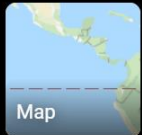


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Isla Fernandina

E5
Isla Santa Cruz
Puerto Ayora

Isla Isabela

Puerto Villamil

Puerto
Baquerizo
Moreno



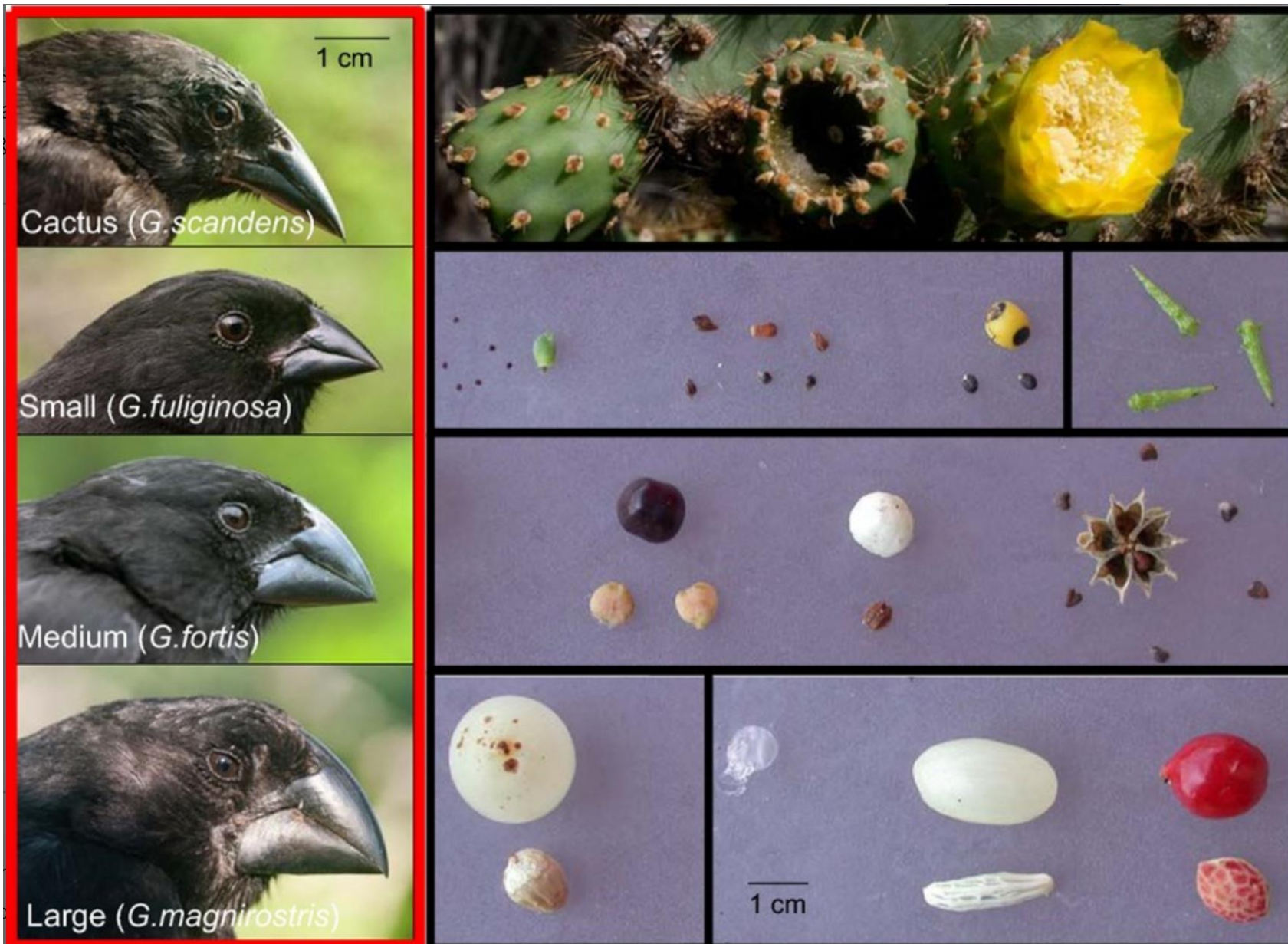
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Darwin's finches and their diet niches: the sympatric coexistence of imperfect generalists

<https://doi.org/10.1111/jeb.12383>



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G. fortis



Big bird



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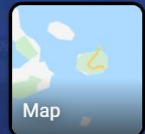
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Puerto Villamil

Puerto
Baquerizo
Moreno



G. scandens



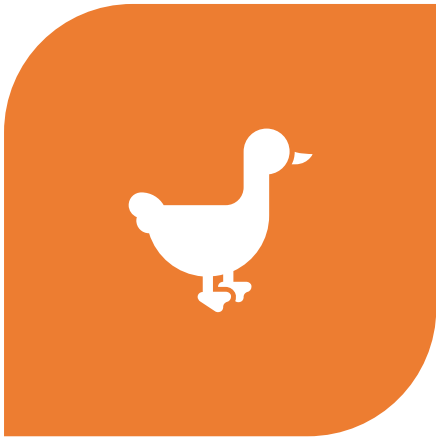
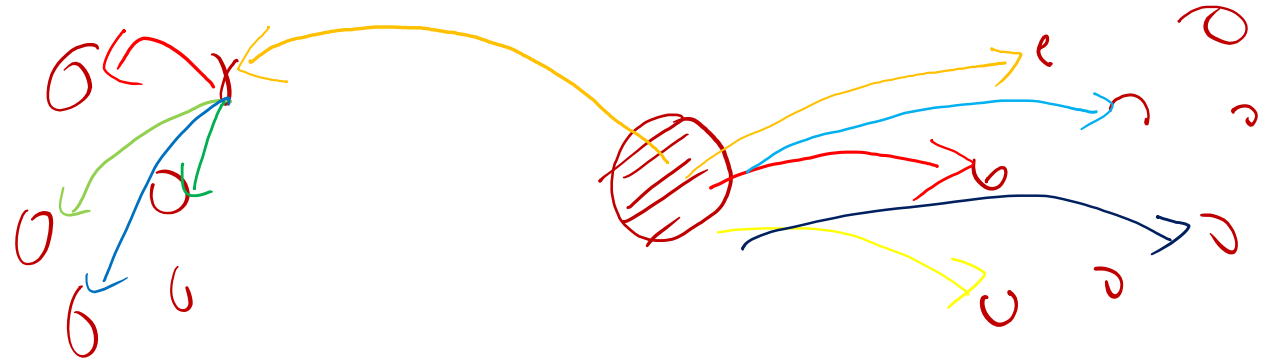
Map

Google



2D





ACTUAL ORIGIN OF THE FOUNDER BIRD?

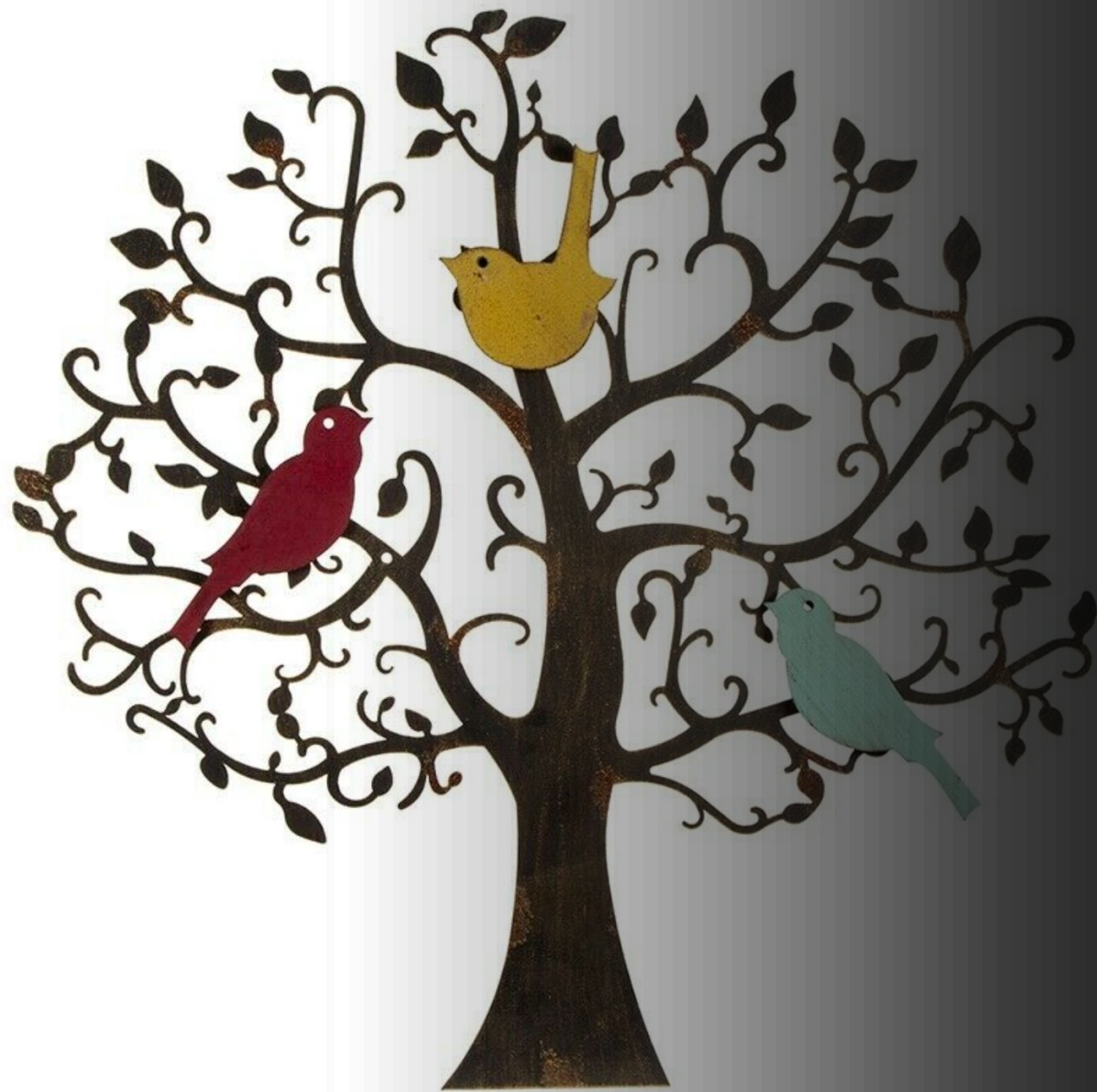


DIFFERENT BIRDS INHABITING DIFFERENT
ISLANDS ORIGINATED FROM SAME
ANCESTOR? – OR –



DIFFERENT BIRDS IMMIGRATED TO THE
ARCHIPELAGO FROM THE MAINLAND AND
GOT DISTRIBUTED TO DIFFERENT ISLANDS?





Phylogenetics



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